Raphael Dedes

CS-340

Professor Scranton

10/18/2026

README

About the Project/Project Title

This project is an interactive dashboard created for Grazioso Salvare. The dashboard uses animal shelter data to help identify dogs that may be suitable for different rescue training programs. Users can filter the data based on rescue type and view the results in a table, a geolocation map, and a pie chart.

Motivation

The goal of this project was to use Python, MongoDB, and Dash to create a dashboard that makes shelter animal data easier to view and understand. The interactive filters help users narrow down the available animals based on the breed, sex, and age requirements for different rescue programs. The map and pie chart provide additional ways to explore the filtered results.

Getting Started

This dashboard project uses Python and MongoDB to retrieve and display animal shelter records.

The following tools and libraries are used:

-Python
-MongoDB
-PyMongo
-JupyterDash and Dash
-Pandas
-Plotly
-Dash Leaflet

Python was used to connect the different parts of the fashboard and apply the rescue filtering criteria. MongoDB stores the animal shelter records, and the existing PyMongo CRUD module provides access to that data. Pandas makes it easier to organize the records, while Dash and JupyterDash provide the interactive dashboard interface. Plotly creates the pie chart, and Dash Leaflet displays the animal locations on the map.

To run the project, first make sure MongoDB is running and the aac.animals collection contains the animal shelter data. Open the dashboard notebook in the appropriate Jupyter or Codio environment, confirm the database connection settings, and run the notebook cells in order.

The CRUD_Python_Module.py file and the Grazioso Salvare Logo.png image must also be available in their expected project locations.

During development, I encountered callback errors while connecting the data table to the map and pie chart. I resolved these issues by checking the callback functions, making sure the filtered data was defined before it was used, and adding in checks for empty results. I also tested the rescue filters and visualizations to confirm that the dashboard responded correctly to user selections.

Usage

1) Run the dashboard notebook to start the application
2) Select All Animals to view the full dataset
3) Select Water Rescue, Mountain or Wilderness Rescue, or Disaster or Individual Tracking to filter the animals
4) Use the data table to search, sort, navigate pages, and select an animal
5) View the selected animal's location on the map
6) Use the pie chart to see the most common breeds in the displayed results
7) Select All Animals to return to the full dataset

Code Example

The following example shows how the dashboard filters animals for the Water Rescue program based on breed, sex, and age.

```
#display all animals
    if filter_type == 'Reset':
        filtered_df = df


    #water rescue
    elif filter_type == 'Water Rescue':
        filtered_df = df[
```

```
    (df['breed'].str.contains(
        r'Labrador Retriever|Chesapeake Bay Retriever|Newfoundland',
        case=False, na=False
    )) &
    (df['sex_upon_outcome'] == 'Intact Female') &
    (df['age_upon_outcome_in_weeks'].between(26, 156))
]
```

This code selects animals whose breed matches one of the preferred Water Rescue breeds, whose sex is Intact Female, and whose age is between 26 and 156 weeks.

Tests

The dashboard was tested by selecting each rescue category and checking that the data table updated to display the appropriate and correct results. The map and pie chart were also checked to confirm that they displayed data without any callback errors.

The All Animals option was tested to confirm that users could return to the full dataset by selecting it.

Screenshots



CS-340 Dashboard

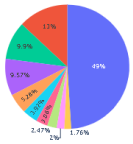
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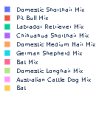
Select Rescue Type:

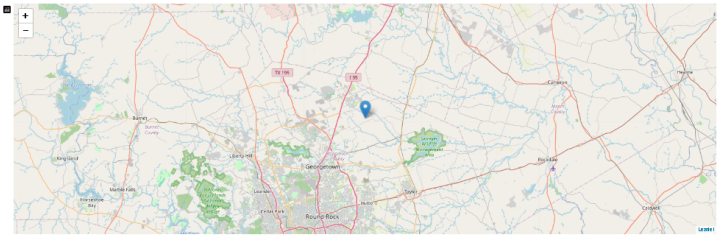
All Animals

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	Filter data...															
○	3	5 years	A736330	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-12	2015-12-22 12:43:00	2015-12-22T12:43:00	Frank		Adoption	Neutered Male	30.7595745111548	-97.9313726387133	156.11144627381
○	2	1 year	A750117	Cat	Domestic Shorthair Mix	Silver Tabby	2015-03-02	2016-03-03 10:58:00	2016-03-03T10:58:00		SCRF	Transfer	Spayed Female	30.6919684968728	-97.7410661874644	52.021927777778
○	4	7 months	A733653	Cat	Siamese Mix	Seal Point	2016-01-25	2016-03-27 12:11:00	2016-03-27T12:11:00	Kitty		Adoption	Intact Female	30.3228461374357	-97.7366374748391	30.821916081250
○	5	2 years	A691824	Dog	Labrador Retriever Mix	Tan/White	2013-11-06	2015-03-30 13:42:00	2015-03-30T13:42:00	Luke		Return to Owner	Neutered Male	30.7.2842159128133	-97.9023874.35286	133.69397142857 1
○	6	5 years	A696884	Dog	Cardigan Welsh Corgi Mix	Sable/White	2010-01-27	2015-01-31 18:30:00	2015-01-31T18:30:00	Lucy	Rabies Risk	Euthanasia	Spayed Female	30.47.37.3658934231	-97.74747.1929467	261.4613925571 43
○	7	2 years	A673130	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-13 15:15:00	2014-03-13T15:15:00	Zach	Aggressive	Euthanasia	Neutered Male	30.2156150323441	-97.1130642112436	100.00729238002
○	1	3 years	A746074	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11T09:00:00		SCRF	Transfer	Neutered Male	30.5000597327 39459	-97.3.6.6.0.7.3.8.7.2.2.3.3.5	156.7.6.7.3.5.7.1.4.2.8.5.7
○	8	1 year	A736951	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 12:00:00	2016-11-27T12:00:00	Max		Adoption	Spayed Female	30.4443112236122	-97.7.3.5.0.0.0.9.3.7.0.3	81.0.6.4.2.8.5.7.1.4.2.8.5.7
○	10	3 months	A664.300	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-02 14:52:00	2013-12-02T14:52:00	Taylor		Adoption	Spayed Female	30.7.8231.0.4.5.1.1.6.2.8.4.1	-97.6.1.2.3.9.7.1.6.2.8.4.6	14.4.0.2.9.6.2.3.8.1.5.2.7.3
○	12	1 year	A.6.6.8.2.6.3	Dog	Pit Bull Mix	Brown/White	2014-02-12 17:36:00	2014-02-12T17:36:00	2014-02-12T17:36:00	Sherlock	Partner	Transfer	Neutered Male	30.4.5.1.5.4.0.2.0.7.3.0.6	-97.4.7.4.1.0.4.5.1.8.6.2.5	62.3.4.0.4.2.5.7.1.4.2.8.6

Top 10 Breeds in Filtered Results



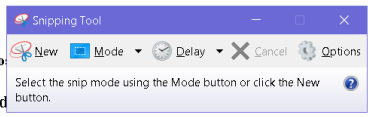






CS-340 Dashbo

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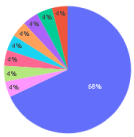


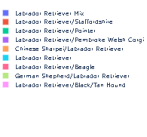
Select Rescue Type:

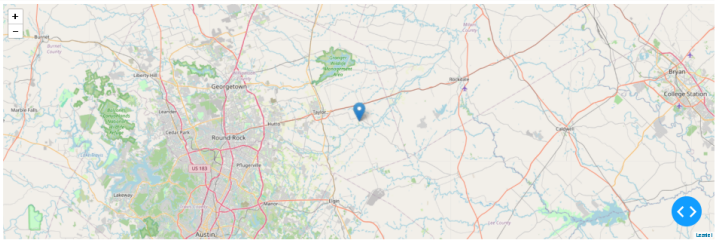
Water Rescue

	rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex_upon_outcome	location_lat	location_long	age_upon_outcome_in_weeks
	Filter data...															
○	30	6 months	A740053	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Euthanasia	Intact Female	30.5420692302233	-97.3.9.0.9.0.4.8.2.3.5.7	30.1.5.4.6.6.4.2.8.5.7.1.4.3
○	339	9 months	A.6.6.9.1.7.7	Dog	Labrador Retriever/Staffordshire	Black/White	2013-03-16	2013-12-16 14:43:00	2013-12-16T14:43:00	Ally		Adoption	Intact Female	30.3.3.4.9.8.3.6.1.2.3.9	-97.6.4.6.5.4.4.5.7.4.6.9.1	40.3.5.5.7.8.6.3.6.3.0.8.6
○	732	3 years	A.7.4.9.7.2.2	Dog	Labrador Retriever Mix	Tan/White	2015-05-20	2017-07-25 14:55:00	2017-07-25T14:55:00	Valentine		Return to Owner	Intact Female	30.4.1.2.3.1.3.4.6.3.0.7.5.7	-97.7.7.2.1.5.4.2.5.7.6.9.3	114.4.8.2.2.8.5.7.1.4.2.8.5.7
○	865	1 year	A.6.7.3.9.6.5	Dog	Labrador Retriever/Pointer	Chocolate/White	2013-02-02	2014-02-07 15:00:00	2014-02-07T15:00:00	Hazel	Partner	Transfer	Intact Female	30.7.3.9.4.5.1.4.5.9.4.4.5	-97.6.9.6.2.3.8.1.6.7.9.1.1	52.6.4.6.4.2.8.5.7.1.4.2.8.6
○	990	1 year	A.7.1.5.3.5.7	Dog	Labrador Retriever/Pembroke Welsh Corgi	Black	2014-11-04	2015-11-05 00:00:00	2015-11-05T00:00:00	Roxy	Partner	Transfer	Intact Female	30.3.9.7.3.2.4.4.3.9.3.3.3.1	-97.2.9.6.5.4.6.1.6.7.2.7	53.7.1.4.2.8.5.7.1.4.2.8.5.7
○	1121	1 year	A.7.9.7.1.5.2	Dog	Labrador Retriever Mix	White/Black	2016-05-30	2017-05-31 14:12:00	2017-05-31T14:12:00	Pirate		Return to Owner	Intact Female	30.3.9.7.1.1.6.5.9.7.9.6.1	-97.5.3.6.3.2.4.3.6.3.5.7.8	52.3.7.0.2.8.8.9.2.2.8.1
○	1625	6 months	A.7.4.0.4.7.1	Dog	Labrador Retriever Mix	Tan/White	2016-12-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Alba		Adoption	Intact Female	30.7.9.6.9.4.3.6.3.2.3.4.1	-97.7.3.2.5.4.5.3.7.6.6.9.4	40.1.4.5.2.8.5.7.1.4.2.8.5.7
○	1747	7 months	A.7.4.3.7.4.7	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-01-14 15:30:00	2017-01-14T15:30:00	Harley		Return to Owner	Intact Female	30.4.2.8.6.6.7.5.4.8.7.3.2.6	-97.4.2.8.8.0.7.1.6.7.3.8.2	33.1.5.4.1.9.6.8.5.1.3.7
○	1.8.8.3	1 year	A.7.6.7.3.3.1	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:00:00	2017-12-03T13:00:00		Partner	Transfer	Intact Female	30.2.8.4.4.1.1.1.6.2.8.3	-97.6.8.8.0.5.4.2.1.0.4.7	53.0.7.3.7.2.8.5.7.1.4.2.8.5
○	2.0.4.1	5 years	A.7.4.7.5.4.6	Dog	Labrador Retriever Mix	Black	2013-05-23	2015-05-23 11:45:00	2015-05-23T11:45:00	Abigail		Return to Owner	Intact Female	30.7.1.3.7.4.4.1.6.9.6.3.1	-97.4.9.3.3.6.4.4.2.9.4.7.3	104.1.5.9.6.5.4.7.1.4.2.8.5

Top 10 Breeds in Filtered Results

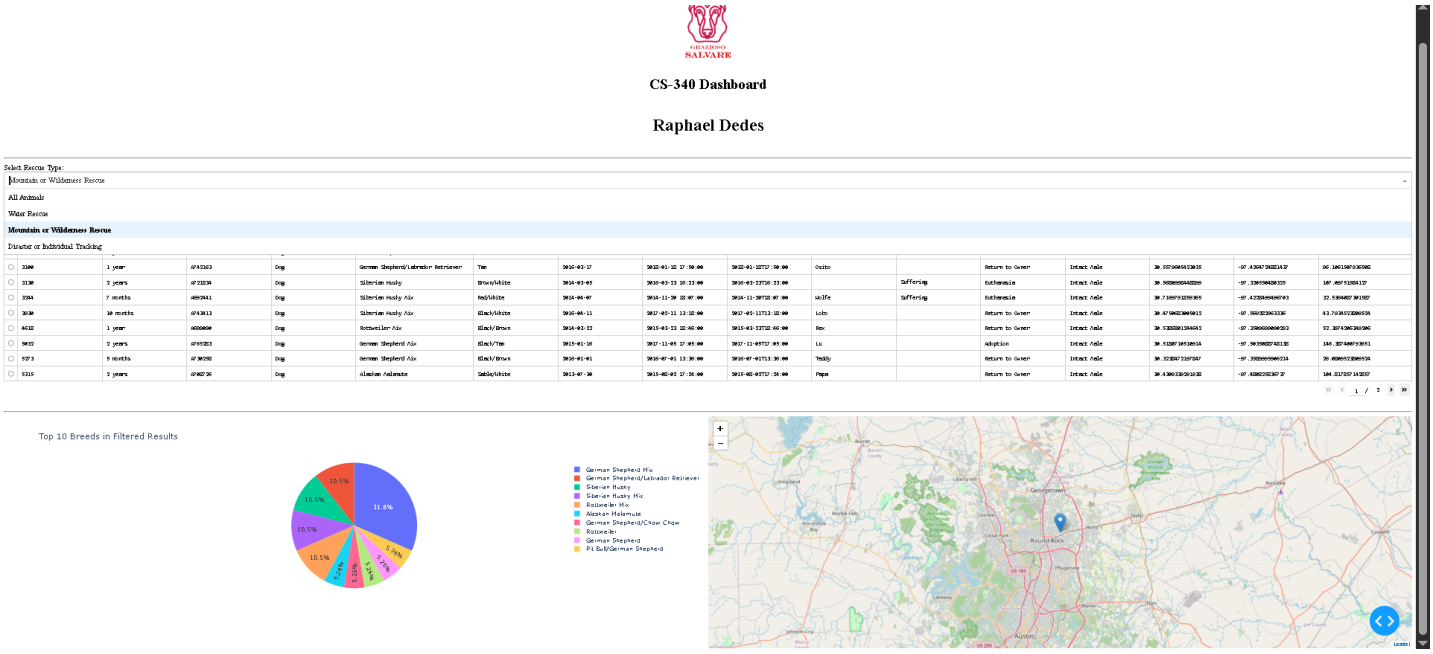






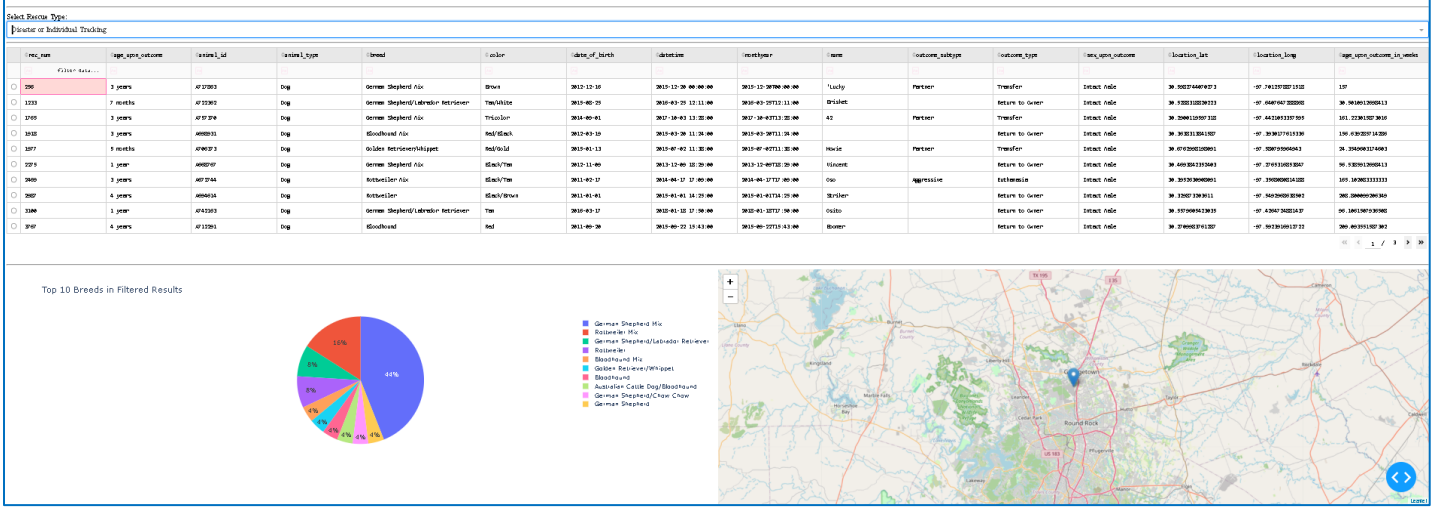
CS-340 Dashboard

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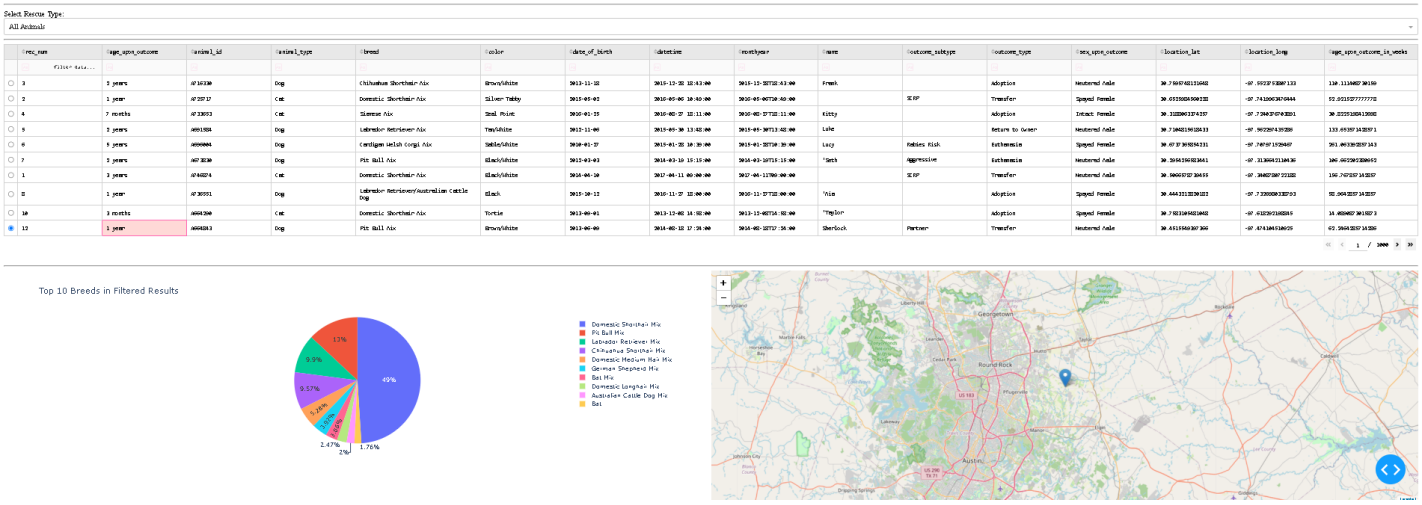
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Contact

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